

Exp No:16

Code:

```
#include <reg51.h>
```

```
#include <stdio.h>
```

```
char ch;
```

```
void serial_ISR(void) interrupt 4
```

```
{
```

```
    if (RI)
```

```
    {
```

```
        RI = 0;          // Clear receive interrupt flag
```

```
        ch = SBUF;        // Read received character
```

```
        printf("\n>> INTERRUPT RECEIVED: ");
```

```
        printf("%c\r\n", ch);
```

```
    }
```

```
}
```

```
void delay(void)
```

```
{
```

```
    unsigned int i;
```

```
    for (i = 0; i < 50000; i++);
```

```
}
```

```
void main(void)
```

```
{
```

```
    SCON = 0x50;          // UART Mode 1, 8-bit, receiver enabled
```

```
    TMOD = 0x20;          // Timer 1, Mode 2
```

```

TH1 = 0xFD;           // 9600 baud
TL1 = 0xFD;

TR1 = 1;              // Start Timer 1
TI = 0;               // EA = 1, ES = 1
IE = 0x90;           // Enable printf transmission

```

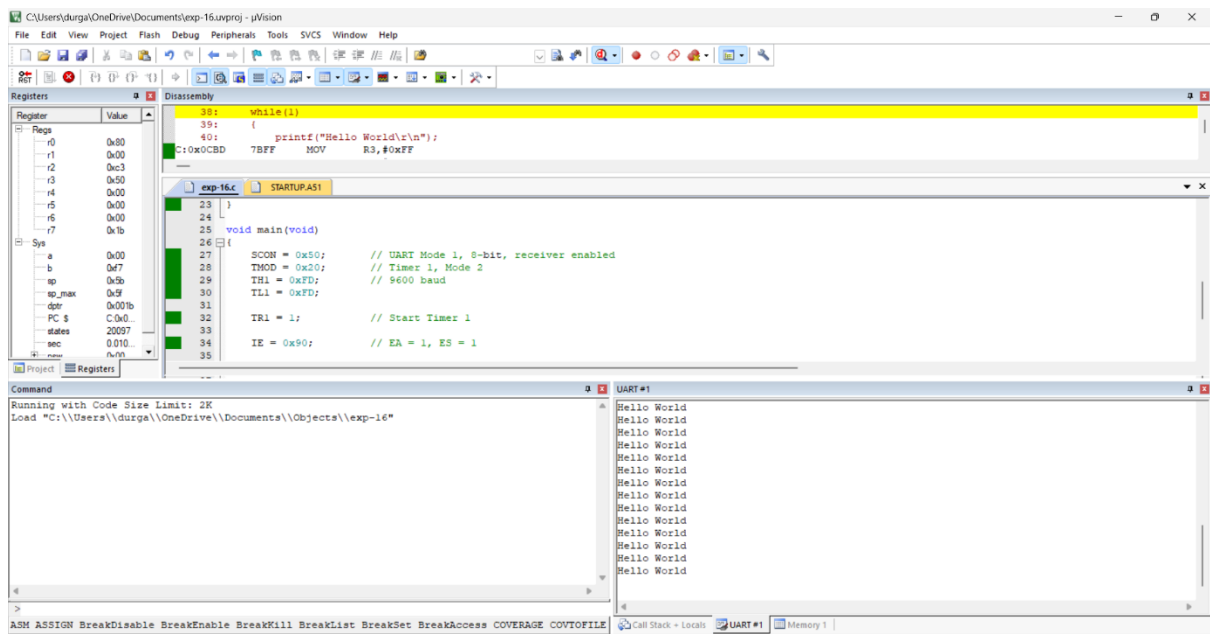
```

while (1)
{
    printf("Hello World\r\n");
    delay();
}

```

Output:

Displaying a Message:



During interruption:

